Summary for conservation law of the form 9.11

$$a(z)\,z_x + z_y = 0 \qquad z(x,0) = f(x)$$

sol'n is given implicitly by

$$z = f(x - a(z)y)$$

characteristics : $x - x_0 = a(z_0)y$

& on charact. $z = f(x_0)$

condition for $z = f(x,y)$ (function)

$$1 + f'(x - a(z)y)\,a'(z)\,y > 0$$

If $= 0$, then $z_x, z_y \to \pm\infty$ & $z \notin C^1$

& sol'n develops a shock.

ex) $z z_x + z_y = 0$ $\qquad$ $a(z) z_x + z_y = 0$

$z(x,0) = -x$ $\qquad$ $z(x,0) = f(x)$

<u>sol'n</u> $z = f(x - a(z) y)$

① Identify $f(x)$ + $a(z)$

$f(x) = -x \qquad a(z) = z$

<u>sol'n</u> $z = f(x - zy)$

$= -(x - zy) = zy - x$

② solve for z: $\quad z - yz = -x$

$z(1-y) = -x \qquad z = \dfrac{-x}{1-y}$

$z = \dfrac{x}{y-1}$

③ ~~②~~ check

$x(y-1)^{-1}$

<u>check soln</u> $\quad z_x = \frac{1}{y-1} \qquad z_y = -\frac{x}{(y-1)^2}$

<u>eq'n</u> $\quad z z_x + z_y = \left(\frac{x}{y-1}\right)\left(\frac{1}{y-1}\right) + \frac{-x}{(y-1)^2} = 0$ ☺

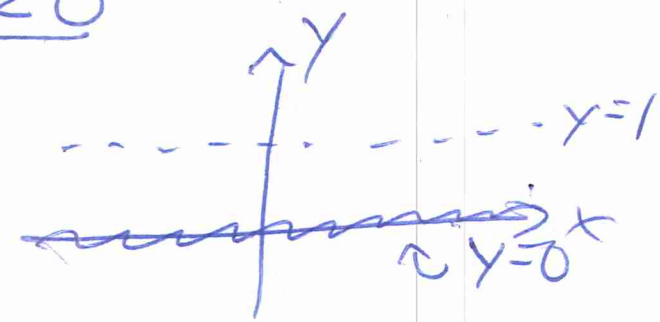
<u>IC</u> $\quad z(x,0) = -x$

$z(x,0) = \frac{x}{-1} = -x$ ✓

what is the domain of soln $z = \frac{x}{y-1}$

$y-1 \neq 0 \qquad y-1 > 0 \qquad \underline{y-1<0}$

I.C. $\quad y = 0 \Rightarrow y < 1$



characteristics (lines)

$x - x_0 = a(z_0) \cdot y$ on charact. $z = f(x_0)$ constant

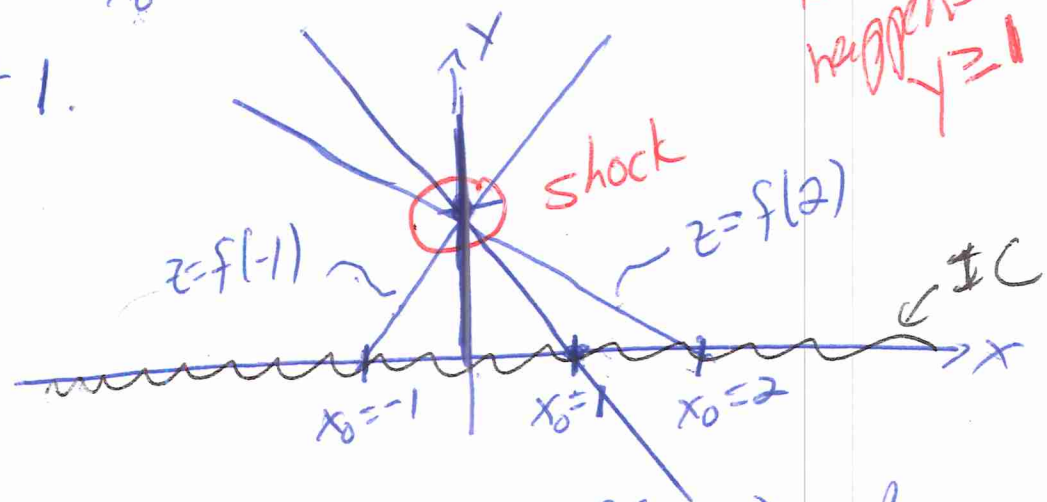
$a(z_0) = z_0 = -x_0$

from I.C.

$x - x_0 = -x_0 y \Rightarrow y = -\frac{1}{x_0} x + 1$

all have a y-intercept at 1.

+ we don't know what happens $y \geq 1$



$x_0 = 1 \quad y = -x + 1$

$x_0 = 0 \quad x = 0$

$x_0 = -1$

$x_0 = 2$

IFT says $z = f(x,y)$ when

$1 + f'(x - a(z)y)\, a'(z)\, y > 0$

$f(x) = -x \quad f'(x) = -1$

$a(z) = z \quad a'(z) = 1$

$1 + (-1)(1)y > 0$

$y < 1$ consistency!

$f(x) = x^2$

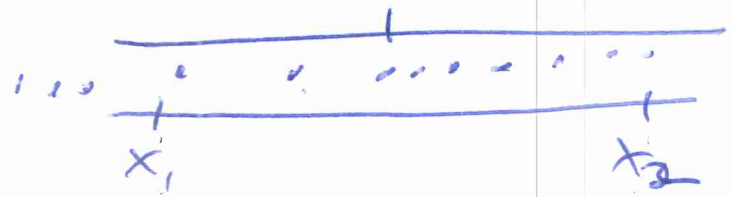
$f'(x) = 2x$

$f'(x - a(z)y) = 2[x - a(z)y]$

Z+T §3.6 Traffic Flow

Treat cars as fluid particles

$\rho(x,t)$ = car density at pt x at time t



$[\rho] = \frac{\text{cars}}{\text{length}}$

$q(x,t)$ = flow rate

$[q] = \frac{\text{cars}}{\text{time}}$

Assume
No stop lights / signs
No entrances / exits

Assume: ρ, q are C^1

Conservation of mass $\quad x_1 < x < x_2$ Not responsible for

$$\underbrace{\frac{d}{dt}\int_{x_1}^{x_2} \rho(x,t)\,dx}_{\text{Total \# of cars}} = \underset{\text{comes in}}{q(x_1,t)} - \underset{\text{goes out}}{q(x_2,t)}$$

$\frac{\text{cars}}{L}$ L

Fund. Thm of Calc

Leibniz rule

$$\int_{x_1}^{x_2} \frac{\partial \rho}{\partial t}\,dx = -\int_{x_1}^{x_2} \frac{\partial q}{\partial x}\,dx$$

$$\int_{x_1}^{x_2} \left(\frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} \right) dx = 0$$

This must hold for all choices of x_1, x_2 where $x_1 < x_2$

<u>Fact</u> If g is C^0 & $\int_I g\,dx = 0$ for all intervals I, Then $g(x) = 0$

$$\Rightarrow \frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} = 0$$

2 unknowns & only one eq'n.
We need one more eq'n

exp. result

$$\boxed{q = c\rho\left(1 - \frac{\rho}{\rho_1}\right)}$$ $\Rightarrow$ another assumption

where ρ_1 = max car density (bumper to bumper)
c = speed limit (5-20 mph above speed limit)

check: $\rho = \rho_1$ $\quad q = c\rho_1(1-1) = 0$

$\rho \ll \rho_1$ $\quad \frac{\rho}{\rho_1} \ll 1$ $\quad q \approx c\rho(1-0)$

$$q = c\rho$$

$\frac{\text{cars}}{\text{time}}$ $\quad \frac{\text{cars}}{\text{Length}}$ $\quad [c] = \frac{\text{Length}}{\text{time}}$ (velocity)

eliminate q

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}\left[c\rho\left(1 - \frac{\rho}{\rho_1}\right)\right] = 0$$

$$\frac{\partial \rho}{\partial t}$$

To be continued...